

Peter W. Fields

Physicist | Machine Learning Researcher | Applied Interdisciplinary Scientist

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EDUCATION

PhD in Physics, University of Chicago – March 2026

Advisor: Stephanie E. Palmer

Bachelor of Science in Physics, The City College of New York – 2019

PUBLICATIONS, PREPRINTS, & SELECTED WRITINGS

Understanding temperature tuning in energy-based models

P. W. Fields, V. Ngampruetikorn, D. J. Schwab, S. E. Palmer.

Preprint (2025). arXiv:2512.09152. *under review at Physical Review Research*.

Code: github.com/peter-fields/temp-tune

Understanding Energy-Based Modeling of Proteins via an Empirically Motivated Minimal Ground Truth Model

P. W. Fields, V. Ngampruetikorn, R. Ranganathan, D. J. Schwab, S. E. Palmer.

Synergy of Scientific and Machine Learning Modeling Workshop at International Conference in Machine Learning (2023). openreview.net/forum?id=vxn5QGPFyi

Presented at ICML 2023 workshop. Code: github.com/peter-fields/toysector

Attention Diagnostics: Testing KL and Susceptibility on the IOI Circuit

peter-fields.github.io/attention-diagnostics/

Why Softmax? A Hypothesis Testing Perspective on Attention Weights

peter-fields.github.io/why-softmax/

SKILLS

Coding: Julia, Python, Mathematica, Unix/CLI, Slurm/HPC clusters, LaTeX, Git

Methods: Information theory; statistical inference; regularization; machine learning. Graduate coursework in Quantum Field Theory, General Relativity, Advanced Statistical Mechanics, Biophysics, Computational Neuroscience

EXPERIENCE

Palmer Theory Group, University of Chicago (Jan 2021 – June 2026)

- Applied information-theoretic and physics-informed methods across projects in protein modeling and biologically motivated neuronal-like systems; developed working fluency in computational neuroscience and quantitative biology.
- Developed synthetic benchmarks and exact maximum likelihood methods to study finite-sample effects and generalization in protein sequence energy-based models.
- Characterized KL divergence structure to understand generative model failure modes.
- Built attention diagnostic pipeline in TransformerLens (Python); tested KL divergence and susceptibility metrics across IOI circuit heads in GPT-2.

Teaching Assistant, General Physics I & II, University of Chicago (2019–2021)

Technical AI Safety Professional Certification – BlueDot Impact, issued May 15, 2026.

Certificate ID: recJ1Yc5DpuIvaC6K